RUIXIN(Jack) SHI

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EDUCATION

Northeastern University, Seattle, WA *May 2024 – Present*
Master of Science in Computer Science
Related courses: Machine Learning, Distributed Systems, Mobile App Development, Assistive Robotics, Advance Software Development

Boston University, Boston, MA *Sep 2020 – May 2024*
Bachelor of Arts in Computer Science
Related courses: Data Structures, Software Engineering, Networks, Artificial Intelligence, Computer Graphics, Operating Systems

PUBLICATIONS

- C. Ming, **R. Shi**, Y. Hu. "NameBERT: Scaling Name-Based Nationality Classification with LLM-Augmented Open Academic Data." **Canadian AI 2026 / PMLR** (Oral Presentation).

TECHNICAL SKILLS

- **Languages:** Python, Java, Kotlin, JavaScript, TypeScript, Go(Golang), Rust, C++, SQL, Swift, Dart, Linux/Unix
- **Frontend & Desktop:** React, Next.js, Electron, Flutter, HTML5/CSS3, Tailwind CSS, Responsive Design
- **Backend & Cloud:** Spring Boot, Node.js, Express, Django, Flask, REST/GraphQL, Microservices, WebSocket, AWS (Lambda, EC2, S3, SES), ByteFaaS, Docker, Kubernetes, CI/CD
- **AI / ML & Agents:** LLM integration, MCP, agent tooling, RAG, NLP, fine-tuning, PyTorch, Transformers, scikit-learn, OpenRouter API, ComfyUI
- **Data & Tools:** MySQL, PostgreSQL, MongoDB, Redis, DynamoDB, Elasticsearch, Git, pytest, JUnit, Jest, Postman, Agile/Scrum, JIRA

EXPERIENCE

TikTok, Software Engineer Intern, *San Jose, CA* *May 2026 – Present*

- Resolved a **long-standing connection-sync defect** between the Android client and the Electron desktop app by re-architecting the link as a **master–slave protocol** — phone broadcasts authoritative state, desktop subscribes as listener — eliminating state divergence **entirely within the existing pipeline, with no added infrastructure**.
- Built an **on-call assistant bot for Lark** on **ByteFaaS** (ByteDance serverless FaaS, AWS Lambda equivalent), grounded in **Lark Docs** runbooks to auto-answer recurring on-call questions and cut repetitive escalations to the on-call engineer.
- Engineered the bot backend for **multi-instance serverless execution** — message **deduplication** and **Redis-backed shared conversation history** so context survives across invocations and instances.

Air Health.co, Software Engineer Lead, *Remote* *Dec 2025 – Jul 2026*

- Led **end-to-end development of a cross-platform native mobile app** in Swift (iOS) and Kotlin (Android), driving feature delivery from requirements through production deployment.
- Oversaw **backend migration from Vercel + Cloudflare + Supabase to AWS** (Lambda, EC2, S3, SES), redesigning the architecture for scalability, reduced cold-start latency, and lower operational cost as user volume grew.
- Integrated a **real-time LLM with on-device text-to-speech (TTS)** pipeline, enabling dynamic voice across iOS/Android with sub-second latency.

Northeastern University, Khoury College of Computer Sciences, Graduate Researcher under Dr. Yifan Hu, *Seattle, WA* *May 2025 – Present*

- **Co-authored and presented (oral)** a paper at **Canadian AI 2026 / PMLR**; built a **1.4M-record pipeline** from the Open Academic Graph and fine-tuned **multilingual BERT (NameBERT)**, outperforming SOTA baselines (NamePrism, EthnicSeer, Name2Nat).
- Designed an **LLM-based augmentation pipeline** for long-tail countries (**~17%** tail Macro-F1 gain); benchmarked vs. GPT-5 / Claude / Gemini at **~400x throughput** (1,831 names/sec) and near-zero per-query cost.

Framo Media L.L.C., AI Development Intern, *Seattle, WA* *Sep 2025 – Dec 2025*

- Integrated **AI agent tools** (ComfyUI, OpenRouter API) into production media workflows, automating multi-step content generation; deployed and administered a **PostgreSQL** server for multi-user asset collaboration.

PROJECT EXPERIENCE

Capstone Showcase Website, *React, TypeScript, Sanity.io, Vercel* *Sep 2025 – Dec 2025*

- Co-built and deployed a **full-stack web platform** for Northeastern capstone projects — **React + TypeScript** frontend, **Sanity.io** headless CMS, **Vercel CI/CD**, multi-user content management.

UnWire — Mobile App for Local Network Chat, *Flutter, Dart* *Sep 2024 – Dec 2024*

- Built a **Flutter-based mobile app** enabling real-time peer-to-peer chat over local networks without internet connectivity; designed the full application architecture and core messaging features end-to-end.

AWARDS

Qualcomm & Microsoft On-Device AI Hackathon — Winner, Team Leader, *Northeastern University* *Mar 2025*

- Built **Fit-Mirror**, a real-time computer-vision system tracking body posture with live projection feedback. Won both **Popular Choice and Judges' Choice** awards. [[GitHub]](fitmirror)